Hannover Re — SEA Climate Risk Reserve Gap | Malaysia & Philippines

Physical (GEV EAL) + Transition (Carbon Pass-Through) + Regulatory Context · R-Ignite · MASA Hackathon 2026

Physical EAL Gap (MYS + PHL)

USD 18.4 M

GEV EAL vs EM-DAT burning cost
MYS +5.9% · PHL +1.3%
PELT regime break: 2007 (both countries)

SEA Transition Pool (3% pass-through)

USD 1.11 bn/yr

NGFS NZ2050 carbon price USD 56/t
MYS USD 0.671bn · PHL USD 0.444bn
Stress 2× scenario: USD 111/t

HRe Share of USD 1.11bn SEA Pool

USD 3.4M – 19M/yr

HRe's proportional slice of SEA transition pool
Floor: 3% SEA alloc · Upper: 10% alloc + 2× stress
8% APAC share · 50% treaty attachment factor

REC 1: Recalibrate Physical EAL

DATA

MYS 100-yr: 216mm | EAL gap +5.9%
PHL 100-yr: 521mm | EAL gap +1.3%
PELT break: 2007 (both markets)

ACTIONS

▸ MYS: +6–8% flood loading on XL layers
▸ PHL: +1–2% floor loading (CI 3× wide)
▸ Replace burning cost with GEV EAL
Source: CHIRPS v2.0 · GEV MLE · PELT 2007

REC 2: Country-Specific Transition Surcharges

DATA

MYS: USD 22.4bn/yr (LULUCF emitter)
PHL: USD 14.8bn/yr (LULUCF net sink)
NGFS NZ2050 USD 56/t | Stress USD 111/t

ACTIONS

▸ MYS: 3–5% surcharge (BNM CCPT C3/C4)
▸ PHL: 1–2% surcharge (Art.6.2 ITMO credit)
▸ Add Climate Warranty Clause at renewal
Source: BNM CCPT 2021 · BSP Circ.1085

REC 3: Pass-Through Governance + ENSO Protocol

DATA

PT 1%→5% = 5× variance:
SEA: USD 0.37–1.86bn/yr
HRe: USD 3.0M–14.9M/yr (10% alloc)
ENSO: $r = -0.016$, $p = 0.93$ — absent (null finding)

ACTIONS

▸ Apply 5% PT if carbon > USD 111/t
▸ ENSO: use NOAA La Niña as TCFD audit flag — NOT a pricing variable
Source: NGFS · NOAA ONI 1950–2026